

STATEMENT OF WORK

1. Objective/Requirements

In support of high-temperature structural instrumentation research for hypersonic vehicles, NASA desires a fiber optic cable capable of withstanding temperatures up to 1,800°F. To achieve measurements up to the desired temperatures, NASA has developed sensors utilizing fiber optic cables that are single mode, silica clad, and gold coated. A current research project requires several hundreds of sensors to be fabricated with gold coated fiber.

Fiber optic cables that are not gold coated are known to become extremely brittle after being exposed to high temperatures, causing failures when running multiple structural load cycles.

Therefore, the purchase request is for single mode fiber optic cable, silica clad (optical fibers with a core of silica glass) with a diameter of 125 micron with an additional gold-coating.

2. Characteristics, Scope, and Specs

Specification of gold-coated fiber:

- Single Mode Core: 9 micron
- Doped Silica Clad: 125 micron (+/- 3 micron)
- Gold-coating: 155 micron
 - No additional coatings besides gold-coating
 - No carbon layers below gold-coating
- Length: 4,000 feet (1,220 meters)

3. Place of Performance

N/A

4. Period of Performance

6-10 weeks from purchase